MIL-B-8952A
19 November 1976

SUPERSEDING
MIL-B-8952(ASG)
22 March 1965

MILITARY SPECIFICATION

BEARING, ROLLER, ROD END, ANTIFRICTION SELF-ALIGNING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers antifriction self-aligning rod end roller bearings operating in the temperature range of -67°F to +300°F (-55° to +149°C) and intermittent spikes to 350°F (177°C).

1.2 Classification. Bearings shall be of the following types, as specified (see 6.2.1).

Type I - External thread rod end bearings

Type II - Internal thread rod end bearings

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-B-197	Bearings, Antifriction, Associated Parts and Subassemblies Packaging of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameters; General Specification For
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, Ohio 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3110

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MS 21220	Bearing, Roller, Rod End Internal Thread, Self-aligning Antifriction, Airframe Heavy Duty, Type II, 067° to 350°F
MS 21221	Bearings, Roller, Rod End, External Thread Self-aligning Antifriction, Airframe Heavy Duty, Type I, -67° to 350°F
MS 21223	Bearings, Roller, Rod End, External Thread Self-aligning Antifriction, Airframe Heavy Duty, Type II, -67° to 350°F
MS 21429	Bearing, Roller, Antifriction, Rod End, External Thread, Self-aligning, Airframe, -67° to 350°F, Type I, Heavy Duty

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

American National Standards Institute

ANSI B46.1-1962 Surface Texture (Surface Roughness, Waviness and Lay)

(Applications for copies of the above publication should be addressed to the American Society of Mechanical Engineers, 345 East 46th Street, New York, New York 10017.)

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National Aerospace Standard

NAS 516 Fitting, Lubrication - 1/8 Inch Drive, Flush Type

(Application for copies of the above publication should be addressed to the National Standards Association, Inc., 1321 Fourteenth St., N. W., Washington, DC, 20005.)

3. REQUIREMENTS

3.1 Qualification. The bearings furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified herein, and which has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Material. The bearings shall conform to MS21220, MS21221, MS21223 and MS21429.

3.3 Design and construction. Only factory new bearings conforming to requirements specified herein shall be furnished.

3.3.1 Hardness. The inner ring and rollers shall be through-hardened and shall have a hardness of Rockwell C 60 to 65. The outer raceway may be through-hardened or case-hardened and shall have a minimum hardness of Rockwell C 60. Total case depth shall not exceed 50 percent of cross sectional area. Rod end shanks shall have a uniform cross sectional hardness with minimum hardness of Rockwell C27.

3.3.2 Surface roughness. The sides of the inner ring shall have a smooth, machined finish that will not exceed 125 roughness height rating (rhr). Otherwise, the surface roughness shall be as specified on the military standard (MS).

3.3.3 Seals. Seals of contact type shall be provided to prevent entrance of dust and to retain the lubricant throughout the temperature range specified. The material shall be reinforced TFE fiberglass.

3.3.4 Lubrication provisions. Provisions for lubrication shall be made by use of removable seals or by use of an external lubrication fitting as specified on the MS. When removable seals are used, they shall be readily removable without damage, and shall be of such construction that they may be reused without undue difficulty. When external lubrication fittings are provided, they shall conform to NAS 516-1 as shown on the applicable drawing.

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3.3.5 Retainers. Roller retainers shall be made of steel or beryllium-copper.

3.3.6 Rings. The outer rings shall be of single piece construction with the rod end shank. The inner ring shall be of single piece construction.

3.3.7 Threads. Threads shall conform to UNJF-3A and UNJF-3B in accordance with MIL-S-8879 for external and internal threads, respectively.

3.3.8 Lubrication. The bearings shall be thoroughly cleaned and dried before being lubricated. The bearings shall be filled with grease conforming to MIL-G-81322 so that at least two-thirds of the bearing void shall be filled. Bearings shall not have been lubricated more than 24 months prior to date of delivery.

3.3.9 Stability. The bearing components shall be dimensionally stable for operation from -67° to $+300^{\circ}$F (-55° to $+149^{\circ}$C).

3.4 Performance

3.4.1 Radial strength

3.4.1.1 Radial limit load rating. The bearing shall have a radial limit load rating of not less than the values specified on the MS.

3.4.1.2 Radial fracture load. The minimum static fracture capacity shall be not less than 1.5 times the radial limit load rating specified on the applicable MS.

3.4.1.3 Radial dynamic load rating. The bearings shall have a radial dynamic load rating of not less than the values specified on the applicable MS. The average life shall be a minimum of 10,000 cycles oscillating through an arc of 90 degrees and shall be tested as specified in 4.5.8.

3.4.2 Axial strength

3.4.2.1 Axial limit load rating. The bearings shall have an axial limit load rating of not less than the values specified on the applicable MS.

3.4.3 Radial play. Radial play, measured before the radial limit load rating test (4.5.6) shall be as specified on the applicable MS.

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3.4.4 Axial play. Axial play, measured before the axial limit load rating test (4.5.9) shall be as specified on the applicable MS.

3.5 Dimensions and tolerances. Dimensions and tolerances shall be in accordance with the applicable MS except as listed herein.

3.5.1 Shank alignment. The point representing the center of self-alignment motion shall be coincident with the axis of the rod end within 0.010 inch.

3.6 Weight. The weight of the bearing shall be as specified on the applicable MS.

3.7 Finish. Except when made of corrosion-resistant steel, the face and outside diameter surfaces, excluding bore, shall be plated. Plating shall be in accordance with QQ-P-416 as specified on the applicable MS. The bore of the bearing shall be coated with the same lubricant as used in the bearing.

3.8 Identification of product. Bearings shall be permanently and legibly marked for identification in accordance with MIL-STD-130. The identification data applied to the bearings shall be as follows.

MS number (on shield or outer race)

Manufacturer's name and trademark. Manufacturer's part number option (on second shield or outer race)

U. S. (on second shield or outer race)

3.9 Workmanship. All details of workmanship shall be high grade to ensure satisfactory service life. Parts shall not contain sharp edges, burrs, loose chips, dirt, or other foreign matter. The mating surfaces of the rod end and the inner ring with rollers shall be free of tool marks, galling, and scratches after assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The examination and testing of the bearings shall be classified as:

a. Qualification (4.3)

b. Quality conformance inspection (4.4).

4.3 Qualification inspection

4.3.1 Test samples shall consist of 22 bearings for each MS dash number, representative of the production bearing. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity (see 6.2).

4.3.2 Test reports. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.3.3 Tests. Qualification inspection shall consist of all the tests specified under 4.5. The tests shall be conducted as specified in table I.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests

b. Sampling tests.

4.4.1 Individual tests. The individual tests shall consist of the examination of product.

4.4.2 Sampling tests. The sampling tests shall consist of the tests of 4.5, except 4.5.8, 4.5.13, and 4.5.14.

4.4.2.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.4.2.1.1 Samples. Sample bearings shall be selected in accordance with inspection level II of MIL-STD-105, acceptable quality level (AQL) of 1.0 percent.

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Table I. Qualification Inspection Samples.

	Sample Number																					
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Examination	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Packaging	X	X	X																			
Hardness	X	X	X																			
Surface Roughness	X	X	X																			
Plating	X	X	X																			
Lubrication	X	X	X																			
Radial Play				X	X	X	X	X	X				X	X	X	X	X	X	X	X	X	X
Axial Play				X	X	X				X	X	X	X	X	X	X	X	X	X	X	X	X
Stabilization				X	X	X																
Seals				X	X	X																
Radial Limit Load							X	X	X													
Radial Fracture Load							X	X	X													
Axial Limit Load										X	X	X										
Axial Fracture Load										X	X	X										
Radial Dynamic Load													X	X	X	X	X	X	X	X	X	X
Magnetic Particle	X	X	X																			

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4.5 Test methods

4.5.1 Examination of product. The bearings shall be carefully examined for conformance to this specification and the applicable MS with respect to dimensions, identification, workmanship, and marking.

4.5.1.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be examined for conformance with section 5.

4.5.2 Hardness. The bearings shall be disassembled and tested for hardness, using a standard Rockwell hardness tester. The test shall be made on the rolling contact area of the inner and outer raceways and on the rollers. Readings for rollers shall be taken on a flat surface. Readings for rod ends shall be taken on or near the end of the shank. The hardness shall be as specified in 3.3.1 otherwise the bearing shall be rejected.

4.5.3 Surface roughness. Surface roughness shall be measured in accordance with ANSI B46.1. The test shall be made on the bearing disassembled in 4.5.2. Any failure to comply with the MS shall be cause for rejection.

4.5.4 Plating. Cadmium plating shall be tested for conformance with 3.7 and the applicable MS. Any failure to conform to the MS shall be cause for rejection.

4.5.5 Lubrication. The lubricant shall be inspected for quality and cleanliness and shall be in conformance with the type of grease specified on the MS.

4.5.6 Radial limit load. The bearings shall be mounted in a rigid support as shown on figure 1, and the bearing rotated manually to determine operational smoothness before testing. The starting friction torque shall be recorded. The appropriate load specified on the applicable MS shall be applied for one minute. The load shall be removed and the bearing inspected for evidence of failure. If starting torque increases by 100 percent or more the bearing shall be considered to have failed.

4.5.7 Radial fracture load. The radial fracture load of 1.5 times the radial limit load specified on the MS shall be applied to the bearing for one minute. After removal of the load, the bearings shall be examined for failure. Bearings that cannot be turned by hand, or that have parts fractured, shall be considered to have failed. Bearings of each MS part number submitted shall be tested.

4.5.8 Radial dynamic load. Ten bearings of each MS part number submitted shall be tested for radial dynamic load. The bearings shall be mounted on a test fixture having suitable support bearings and drive linkages to provide oscillation of the inner ring. The radial dynamic load specified on the MS

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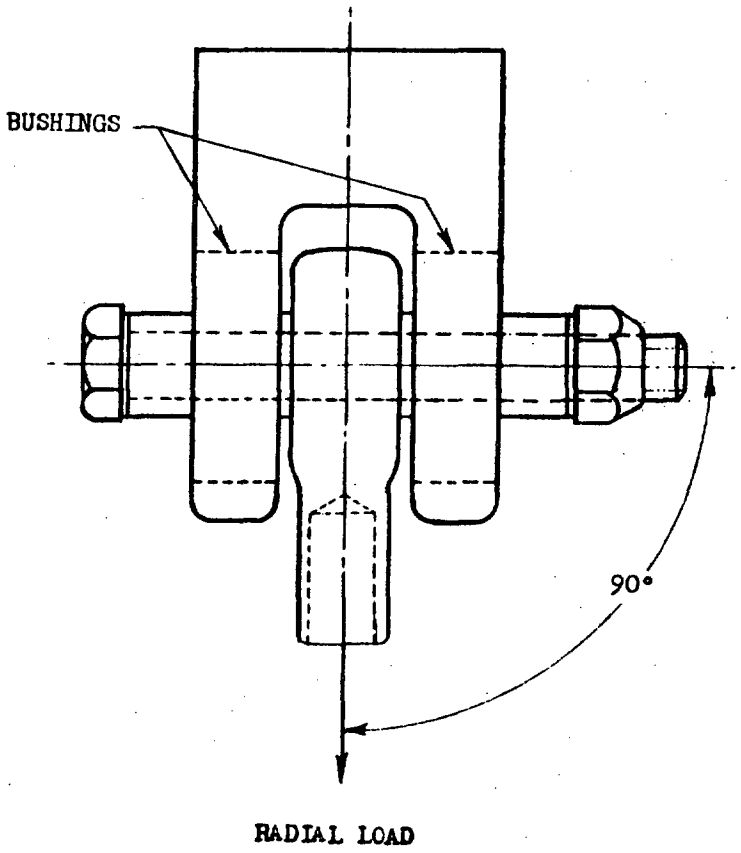


Figure. 1. Radial Strength Test Setup

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shall be applied unidirectionally with respect to the outer ring and maintained constant within ±1 percent. Each bearing shall be oscillated through an arc of 90 degrees and back to the starting position. Termination of each test shall be an indication of spalled race surfaces as measured by roughness of operation or increase in radial internal clearance, either method established to yield about 90 percent failed samples. The bearings shall be disassembled and the raceways inspected. Failure is defined as a miniumum of 0.01 square inches of spalled race surface. Data is to be Weibull plotted for evaluation. If the B-50 life of the bearings is less than 10,000 cycles, the bearings shall be considered to have failed the test.

4.5.9 Axial limit load. The bearing shall be mounted in rigid support as shown on figure 1 and the bearing rotated manually to determine operational smoothness before testing. The starting friction torque shall be measured. The appropriate load specified on the applicable MS shall be applied for one minute. The load shall be removed and the bearing inspected for evidence of failure. If starting torque increases by 100 percent or more, the bearing shall be considered to have failed.

4.5.10 Axial fracture load. The axial fracture load of 1.5 times the axial limit load specified on the MS shall be applied to the bearing for one minute. After removal of the load, the bearings shall be examined for failure. Bearings that cannot be turned by hand, or that have parts fractured, shall be considered to have failed.

4.5.11 Radial play. The bearings of each MS size shall be tested for radial play. The inner ring shall be held rigidly and a radial load of 22 pounds applied alternately in opposite directions to the outer ring. A dial indicator shall be used to measure total movement of the outer ring. The radial movement shall be in conformance with the radial clearance values specified on the applicable MS.

4.5.12 Axial play. Bearings of each MS size shall be tested for axial play. The outer ring shall be held rigidly and an axial load of 22 pounds applied alternately in opposite direction to the inner ring. A dial indicator shall be used to measure total movement of the inner ring. The axial movement shall be in conformance with the axial clearance values specified on the applicable MS.

4.5.13 Seals (qualification). Bearings shall be held at -67°F (-55°C) for 300 hours, then at 300°F (149°C) for 300 hours. While being operated in the test chamber at three revolutions per minute (rpm), the bearings shall be subjected to sand and dust test, method 510.1 of MIL-STD-810. Seal deterioration, nonretention of lubricant, or ingress of dust into the bearings shall be considered seal failure.

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4.5.14 Stabilization (qualification). The bearing shall be held at 300°F (149°C) ±10°F for 100 hours; allowed to return to room temperature, then measured. Bearings that show a dimensional change of more than 0.0001 inch per inch of diameter shall be considered to have failed. The bearings shall then be cold soaked at -67°F (-55°C) ±10°F for 100 hours, allowed to return to room temperature, then measured. Bearings that show a dimensions change of more than 0.0001 inch per inch of diameter from the last reading shall be considered to have failed. All measurements shall be made at room temperature and shall be made at a single reference position.

4.5.15 Magnetic particle inspection. This inspection shall be in accordance with MIL-I-6868. The bearings shall be magnetically inspected by both longitudinal and circular methods. Certification of such inspection shall be furnished with qualification and quality conformance inspection samples.

5. PACKAGING

5.1 Cleaning, preservation, and packaging.

5.1.1 Level A. Cleaning, preservation, and packaging shall be in accordance with level A of MIL-B-197.

5.1.2 Level C. Cleaning, preservation, and packaging shall be in accordance with MIL-B-197.

5.2 Packing

5.2.1 Level A. Bearings, cleaned, preserved, and packaged as specified in 5.1, shall be packed for domestic shipment and storage in accordance with MIL-B-197.

5.2.2 Level B. Bearings, cleaned, preserved, and packaged as specified in 5.1, shall be packed for domestic shipment and storage in accordance with MIL-B-197.

5.2.3 Level C. Bearings, cleaned, preserved, and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. The containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

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5.3 Marking of shipments. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129. The package shall be marked with the lubrication date.

6. NOTES

6.1 Intended use. Sealed and shielded antifriction bearings are intended for use in aircraft control systems. Where the exclusion of dust, salt spray, and other extraneous materials is desirable, contact seals are the preferred type. These bearings are not intended for application involving high rotative speeds.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements

a. Title, number, and date of this specification

b. Type and MS Part No. (see 1.2 and 3.2)

c. Quantity

d. Data requirements (see 3.1 and 4.3.2)

e. Levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data specified in paragraph 4.3.2 will be listed directly on a DD Form 1423 incorporated into the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, ATTN: ENFL, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

Custodians:
Army -
Navy - AS
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 3110-0494

Review activities:
DSA - IS

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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